

A Trivial-Bundle Partial Answer to the C^* -Algebra Ball-Covering Question

Run fa_banach_001, lane 19

June 17, 2026

Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when its pure state space, i.e. K , has a countable π -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

Recall that a family of nonempty open sets in K is a π -basis if every nonempty open set in K contains a member of the family.

Theorem 1. *Let K be a locally compact Hausdorff space and let X be a nonzero Banach space.*

1. *If $C_0(K, X)$ has the ball-covering property, then K has a countable π -basis.*
2. *If K has a countable π -basis and X has the ball-covering property, then $C_0(K, X)$ has the ball-covering property.*
3. *If K has a countable π -basis and X is separable, then $C_0(K, X)$ has the uniform ball-covering property.*

Corollary 1. *Let A be a nonzero separable C^* -algebra. Then $C_0(K, A)$ has BCP/UBCP if and only if K has a countable π -basis.*

This gives a noncommutative extension of the commutative $C_0(K)$ criterion for the large class of trivial continuous C^* -bundles with separable fiber, including $C_0(K, M_n)$ for $n \geq 2$ and $C_0(K, \mathcal{K}(H))$ for separable H .

After this packet was first drafted, a later literature check identified that Liu–Liu–Lu–Zheng [2] already state the broader Banach-valued stability theorem for $C_0(K, X)$, including strong and uniform versions. Accordingly, the packet should be read as a C^* -specialized self-contained proof and application note for the source question, not as a novel theorem beyond that vector-valued result.

Proof Intuition

The obstruction is still topological. If K has no countable π -basis, then the near-norm sets of any countable family of candidate centers cannot probe all open subsets of K . A vector-valued bump supported in an open set missed by those near-norm sets is zero where each candidate center almost attains its norm, so it is outside every admissible ball.

Conversely, a countable π -basis supplies countably many scalar bump functions. Near a point where a vector-valued function f nearly attains its norm, choose a fiber ball center covering that fiber value and multiply it by a small scalar bump supported where f changes little. Outside the support the distance is at most $\|f\|$; inside the support, convexity controls the whole transition from 0 to the fiber center.

Proof

We use the standard equivalent form of BCP: the unit sphere of a normed space Y is covered by countably many open balls $B(y_n, \|y_n\|)$, $y_n \neq 0$. Indeed, any open ball off the origin has radius at most the norm of its center, and enlarging to radius $\|y_n\|$ still keeps the origin out of the open ball.

Lemma 1 (the base obstruction). *Let $X \neq \{0\}$. If K has no countable π -basis, then $C_0(K, X)$ fails BCP.*

Proof. Let (F_n) be any sequence in $C_0(K, X)$. We show that the balls $B(F_n, \|F_n\|)$ cannot cover the unit sphere. For $n, k \geq 1$, set

$$U_{n,k} = \{t \in K : \|F_n(t)\| > \|F_n\| - 1/k\}.$$

These sets are open and nonempty whenever $F_n \neq 0$. Since K has no countable π -basis, there is a nonempty open set $W \subset K$ such that no $U_{n,k}$ is contained in W . Choose $h \in C_c(K)$ with $0 \leq h \leq 1$, $\|h\|_\infty = 1$, and $\text{supp } h \subset W$. Choose $x_0 \in X$ with $\|x_0\| = 1$, and put

$$G(t) = h(t)x_0.$$

Then $G \in C_0(K, X)$ and $\|G\| = 1$.

Fix n . For every k , choose $t_{n,k} \in U_{n,k} \setminus W$. Since $\text{supp } h \subset W$, we have $G(t_{n,k}) = 0$, hence

$$\|G - F_n\| \geq \|F_n(t_{n,k})\| > \|F_n\| - 1/k.$$

Letting $k \rightarrow \infty$ gives $\|G - F_n\| \geq \|F_n\|$. Thus G is in none of the open balls $B(F_n, \|F_n\|)$. Since the sequence (F_n) was arbitrary, $C_0(K, X)$ fails BCP. $\square$

Lemma 2 (lifting a fiber BCP cover). *If K has a countable π -basis and X has BCP, then $C_0(K, X)$ has BCP.*

Proof. Let (V_j) be a countable π -basis of K . By local compactness and regularity, for each j choose $h_j \in C_c(K)$ such that

$$0 \leq h_j \leq 1, \quad \|h_j\|_\infty = 1, \quad \text{supp } h_j \subset V_j.$$

Let $(a_m) \subset X \setminus \{0\}$ be BCP centers for X , so that

$$S_X \subset \bigcup_m B(a_m, \|a_m\|).$$

Discard centers whose balls miss S_X , and replace the remaining centers by $y_m = 4a_m$. The cover property is preserved because if $\|x - a_m\| < \|a_m\|$, then

$$\|x - 4a_m\| \leq \|x - a_m\| + 3\|a_m\| < 4\|a_m\| = \|y_m\|.$$

Also every retained a_m satisfies $\|a_m\| > 1/2$, so $\|y_m\| > 2$.

We claim that the countable family

$$G_{j,m}(t) = h_j(t)y_m$$

is a BCP family for $C_0(K, X)$. Let $f \in C_0(K, X)$ with $\|f\| = 1$. Choose $t_0 \in K$ with $\alpha := \|f(t_0)\| > 0$, and put $x = f(t_0)/\alpha \in S_X$. Choose m such that $\|x - y_m\| < \|y_m\|$. Since $0 \leq \alpha \leq 1$,

$$\|\alpha x - y_m\| \leq \alpha\|x - y_m\| + (1 - \alpha)\|y_m\| < \|y_m\|.$$

Moreover $\|y_m\| > 2$, so both endpoint distances $\|\alpha x\|$ and $\|\alpha x - y_m\|$ are strictly smaller than $\|y_m\|$. By convexity, the function

$$s \mapsto \|\alpha x - sy_m\|, \quad 0 \leq s \leq 1,$$

is bounded by the maximum of its endpoint values, and is therefore strictly less than $\|y_m\|$ on the whole interval.

Choose a neighbourhood U of t_0 so small that

$$\|f(t) - f(t_0)\| < \|y_m\| - \sup_{0 \leq s \leq 1} \|\alpha x - sy_m\| \quad (t \in U).$$

Pick j with $V_j \subset U$. If $t \notin \text{supp } h_j$, then

$$\|f(t) - G_{j,m}(t)\| = \|f(t)\| \leq 1 < \|y_m\|.$$

If $t \in \text{supp } h_j$, then $t \in U$ and

$$\|f(t) - G_{j,m}(t)\| \leq \|f(t) - f(t_0)\| + \|\alpha x - h_j(t)y_m\| < \|y_m\|.$$

Thus $\|f - G_{j,m}\| < \|G_{j,m}\|$. This covers the unit sphere. $\square$

Lemma 3 (separable fibers give UBCP). *If K has a countable π -basis and X is separable, then $C_0(K, X)$ has UBCP.*

Proof. Let (u_m) be a countable $1/16$ -net in S_X , and let (h_j) be the scalar bumps from the preceding proof. Consider the centers

$$G_{j,m}(t) = 4h_j(t)u_m$$

and use the fixed radius $7/2$. These balls are uniformly bounded and each is disjoint from the open ball $B(0, 1/2)$, since $\|G_{j,m}\| = 4$.

Let $f \in S_{C_0(K,X)}$. Choose t_0 with $\alpha = \|f(t_0)\| > 3/4$, set $x = f(t_0)/\alpha$, and choose m with $\|x - u_m\| < 1/16$. Then

$$\|\alpha x - 4u_m\| \leq \alpha\|x - u_m\| + (4 - \alpha) < \frac{1}{16} + \frac{13}{4} < \frac{7}{2}.$$

Also $\|\alpha x\| < 7/2$. By convexity,

$$\sup_{0 \leq s \leq 1} \|\alpha x - 4su_m\| < 7/2.$$

Choose a neighbourhood U of t_0 on which f varies by less than the positive margin between $7/2$ and the displayed supremum, and then choose j with $V_j \subset U$. The same outside-support and inside-support estimates as in Lemma 2 give

$$\|f - G_{j,m}\| < 7/2.$$

Thus the unit sphere is covered by countably many balls of fixed radius $7/2$, all avoiding $B(0, 1/2)$. This is UBCP. $\square$

Theorem 1 follows from Lemmas 1, 2, and 3. Corollary 1 follows because every separable Banach space is a separable fiber in Lemma 3, and Lemma 1 gives the reverse implication for every nonzero fiber.

Scope and Limitations

This is not a full characterization of arbitrary C^* -algebras with BCP/UBCP. It covers the important and natural class of trivial continuous fields $C_0(K, A)$ and shows that, in this class, the commutative base-space condition is exactly the right obstruction. The argument does not address general nontrivial continuous fields, extensions, quotients, or arbitrary nonseparable fibers.

Novelty and Search Bounds

The run indexes were searched for arXiv:2311.14257, ball-covering property, C^* -algebra, pure state space, π -basis, BCP, and UBCP. No existing packet for this target was found. Web searches on June 17, 2026 found no later direct characterization of all C^* -algebras with BCP/UBCP. They did identify arXiv:2111.04921 [2], which already proves the broader Banach-valued $C_0(K, X)$ stability theorem underlying this trivial-bundle subcase, and arXiv:2507.02261 on approximation criteria for operator spaces.

Human Review Focus

The main checks are:

- in Lemma 1, the use of near-norm open sets to turn failure of countable π -basis into a vector-valued bump escaping every countable ball cover;
- in Lemma 2, the convexity estimate through the transition region of the scalar bump;
- in Lemma 3, the fixed constants 4, $7/2$, and $1/2$ used to obtain UBCP.

References

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Minzeng Liu, Rui Liu, Jimeng Lu, and Bentuo Zheng, *Ball covering property from commutative function spaces to non-commutative spaces of operators*, arXiv:2111.04921, 2021.